

Naziru Halilu

Digital Agriculture · Computer Vision · Geospatial Machine Learning

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SUMMARY

Agricultural engineer with a long-term research focus on the digitalization of agriculture, spanning computer vision, geospatial machine learning, agricultural control systems, and high-throughput phenotyping, grounded in a strong foundation in agricultural machinery, mechanization, and farm power systems. First Class Honours B.Eng (CGPA 4.77/5.00, Top 1%), currently an Erasmus Mundus Joint Master's scholar.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD), Sustainable Agriculture (SUSTAGRI) 2025 – 2027

Public University of Navarre (Spain) · UTAD (Portugal) · Agricultural University of Athens (Greece)

B.Eng, Agricultural & Bioresources Engineering 2019 – 2024

Ahmadu Bello University, Zaria, Nigeria · First Class Honours · CGPA 4.77/5.00, Top 1%

National Diploma, Agricultural Engineering Technology 2016 – 2018

Samaru College of Agriculture, ABU · Distinction · CGPA 3.78/4.00, Top 1%

RESEARCH & PROFESSIONAL EXPERIENCE

EMJMD Research Scholar — Agricultural University of Athens, Greece Sep 2026 – Jan 2027

- Continued graduate-level research training in smart design and management of irrigation systems.
- Built NaCROP, a modular Python/Tkinter pipeline for evapotranspiration, soil-water balance, irrigation scheduling, and a 4-step crop-growth simulation for five smallholder crops around Zaria, Nigeria.

Summer Intern, Computer Vision Group — Dept. of Computer Engineering, ABU, Zaria Jun – Aug 2026

- Extended the Geometry-Agnostic Contrastive Learning (GACL) framework with four new geometry-invariant components, reaching 64.4% classification accuracy (macro AUC 0.882 vs 5.3% chance) across 19 disease classes on 1,543 field images; first-author paper under review at Biosystems Engineering (Elsevier).

EMJMD Research Scholar — UTAD, Portugal Feb – Jun 2026

- Integrated GIS, remote sensing, and ML pipelines across 65+ hectares of Portuguese trial sites (3 cropping systems), R^2 0.978–0.995, cutting lime demand by 2.69 tonnes via variable-rate application.
- Developed Multitask Singularity Regression Kriging (MSRK), reducing residual kriging uncertainty ~20% across 17,864 samples on 16.85 ha (Moran's I 0.72–0.98, $p < 0.001$).
- Built NDVI/logarithmic regression harvest-date forecasting ($R^2 = 0.9784$) for a 6-ha UTAD vineyard.
- Ran terrain characterisation and a 25-trap monitoring network for grape moth across a 42.97-ha Douro vineyard estate; costed a pheromone diffuser programme (EUR 2,670.6–3,052.1).

EMJMD Research Scholar — Public University of Navarre, Spain Sep 2025 – 31 Jan 2026

- Built a 14-module open-source pipeline for RGB/multispectral vegetation indexing and CNN/Faster R-CNN inference, 33.8% balanced accuracy vs 4.55% chance baseline.
- Designed STADOS, a five-stage radiometric harmonization framework; cut temporal NDVI variance 39%/31% on Sentinel-2 imagery ($R^2 \geq 0.999$ fidelity).
- Built a GIS/regression-kriging pipeline (550 obs., 14.35 km², Tudela, Spain) for livestock slurry suitability; RMSE = 7.78, $R^2 = 0.77$.
- Built a Bayesian-grounded digital twin (hold-out $R^2 = 0.90$) for gully erosion biocontrol; 47.8% reduction in annual soil loss, NSE 0.88–0.94.

Undergraduate Project Assistantships — ABU, Nigeria Mar 2024 – Mar 2025

- Data collection, statistical analysis, and literature review across concurrent undergraduate research projects in water/irrigation systems, soil/erosion assessment, and postharvest/animal systems.

Project Assistant (Remote Data Lead), Smart Greenhouse Automation — IAR-ABU, Nigeria 2024 – 2025

- Engineered adaptive irrigation-scheduling algorithms fusing real-time soil-moisture and microclimate telemetry for automated greenhouse control.

Student Research Assistant, Precision Crop Systems — NAERLS, Nigeria

2022 – 2024

- Trained/validated CNN crop-stress detection models (78% accuracy on multispectral UAV imagery); automated UAV flight-path planning.

Assistant Farm Manager (part-time) — Yankari Nasarawa Farms, Nigeria

2022 – 2024

- Managed automated climate-control and water-dosing systems for a commercial poultry facility.

Project Assistant, Smart Irrigation & Water Systems — BOMO-Basin Project, Nigeria

2021 – 2022

- Implemented closed-loop control logic for a 1-hectare smart drip-irrigation array.

Research Intern — NCAM, Ilorin & IAR, Zaria, Nigeria

2017 – 2020

- Built Python data-logging pipelines to benchmark mechanization efficiency; supported baseline hydrological research.

PUBLICATIONS

- Halilu, N., & Sulaiman, J. (2026). Real-Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. *IJRSI*, 13(7), 2951-2989.
- Halilu, N. (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. *IJRSI*, 13(7), 2920-2950.
- Halilu, N., & Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food-Quality Assessment. *IJRSI*, 13(7), 2990-3020.
- Ismail, K., & Halilu, N. (2024). AI-Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. *Proc. AfCPA*, 2(1), 100-120.
- Halilu, N., & Sulaiman, J. Geometry-Agnostic Contrastive Learning (GACL). *Biosystems Engineering (Elsevier)*, Under Review.
- Halilu, N. (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (*Ipomoea* spp.). *Environmental Monitoring and Assessment (Springer Nature)*, Under Review.
- Halilu, N., & Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. *European Journal of Agronomy (Elsevier)*, Under Review.

TECHNICAL SKILLS

Machine Learning & AI:	PyTorch, TensorFlow, Keras, scikit-learn, CNNs
Geospatial AI / GIS:	ENVI, QGIS, ArcGIS, Google Earth Engine, Sentinel imagery
Crop & Environmental Modelling:	APSIM Next Generation, crop simulation
Programming & Data Science:	Python, MATLAB, R, Pandas, NumPy, Git/GitHub, Zenodo

AWARDS & HONORS

Erasmus Mundus Joint Master Scholarship (2025) · Best Graduating Student, B.Eng (2024) · Nigerian Federal Government Award for Academic Excellence (2024) · Best Undergraduate Research Paper Award, ABU (2024) · Best Overall Graduating Student, National Diploma (2019) · PTDF Undergraduate Fellow (2023)

PROFESSIONAL MEMBERSHIPS

ASABE · Nigerian Institution of Agricultural Engineers (NIAE) · Nigerian Society of Engineers (NSE) · Pupil Engineer, COREN